SOURCE: MEPC.259(68).pdf
resolution MEPC.184(59), the 2009 Guidelines for exhaust gas cleaning systems (hereinafter referred to as "2009 EGCS Guidelines"),

NOTING that the revised MARPOL Annex VI entered into force on 1 July 2010,

NOTING ALSO that regulation 4 of MARPOL Annex VI allows the use of an alternative compliance method at least as effective in terms of emission reductions as that required by MARPOL Annex VI, including any of the standards set forth in regulation 14, taking into account guidelines developed by the Organization,

RECOGNIZING the need to update the 2009 EGCS Guidelines accordingly,

HAVING CONSIDERED, at its sixty-eighth session, draft amendments to the 2009 EGCS Guidelines, prepared by the Sub-Committee on Pollution Prevention and Response, at its second session,

1 ADOPTS the 2015 Guidelines for exhaust gas cleaning systems , as set out in the annex to the present resolution;

2 INVITES Administrations to take these Guidelines into account when allowing the use of an exhaust gas cleaning system in accordance with regulation 4 of MARPOL Annex VI;

3 REQUESTS Parties to MARPOL Annex VI and other Member Governments to bring

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use of an exhaust gas cleaning system in accordance with regulation 4 of MARPOL Annex VI;

3 REQUESTS Parties to MARPOL Annex VI and other Member Governments to bring these Guidelines to the attention of shipowners, ship operators, shipbuilders, marine diesel engine manufacturers and any other interested groups;

4 INVITES Administrations to provide for collection of data as described in appendix 3 of these Guidelines;

5 AGREES to keep these Guidelines under review in the light of experience gained with their application;

6 SUPERSEDES the 2009 EGCS Guidelines adopted by resolution MEPC.184(59).
RESOLUTION MEPC.259(68)
(adopted on 15 May 2015)
2015 GUIDELINES FOR EXHAUST GAS CLEANING SYSTEMSMEPC 68/21/Add.1
Annex 1, page 2

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
ANNEX

2015 GUIDELINES FOR EXHAUST GAS CLEANING SYSTEMS

1 INTRODUCTION

1.1 Regulation 14 of Annex VI requires ships to use fuel oil with a sulphur content not exceeding that stipulated in regulations 14.1 or 14.4. Regulation 4 allows, with the approval of the Administration, the use of an alternative compliance method at least as effective in

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exceeding that stipulated in regulations 14.1 or 14.4. Regulation 4 allows, with the approval of the Administration, the use of an alternative compliance method at least as effective in terms of emission reductions as that required by the Annex, including the standards set forth in regulation 14. The Administration of a Party should take into account any relevant guidelines developed by the Organization pertaining to alternatives provided for in regulation 4.

1.2 Similar to a NO X emission reduction system, an exhaust gas cleaning (EGC) unit may be approved subject to periodic parameter and emission checks or the system may be equipped with a continuous emission monitoring system. These guidelines have been developed with the intention of being objective and performance oriented. Furthermore, use of the SO 2(ppm)/CO2(%) ratio method will simplify the monitoring of SO X emission and facilitate approval of an EGC unit. See appendix II for the rationale explaining the use of SO2(ppm)/CO2(%) as the basis for system monitoring.

1.3 Compliance should be demonstrated on the basis of the SO 2(ppm)/CO2(% v/v) ratio values.

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facilitate approval of an EGC unit. See appendix II for the rationale explaining the use of SO2(ppm)/CO2(%) as the basis for system monitoring.

1.3 Compliance should be demonstrated on the basis of the SO 2(ppm)/CO2(% v/v) ratio values.

Table 1: Fuel oil sulphur limits recorded in regulations 14.1 and 14.4 and corresponding emissions values

Fuel oil sulphur content (% m/m)	Ratio emission SO2(ppm)/CO2(% v/v)
4.50	195.0
3.50	151.7
1.50	65.0
1.00	43.3
0.50	21.7
0.10	4.3

Note: The use of the ratio emissions limits is only applicable when using petroleum based distillate or residual fuel oils. See appendix II for application of the ratio method.

1.4 These guidelines are recommendatory in nature, however, Administrations are

invited to base the implementation of the relevant requirements of regulation 4 of MARPOL Annex VI on them.

2 GENERAL

2.1 Purpose

2.1.1 The purpose of these guidelines is to specify the requirements for the testing, survey certification and verification of EGC systems under regulation 4 of MARPOL Annex VI to ensure that they provide effective equivalence to the requirements of regulations 14.1 and 14.4 of MARPOL Annex VI.

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SOURCE: MEPC.259(68).pdf
certification and verification of EGC systems under regulation 4 of MARPOL Annex VI to ensure that they provide effective equivalence to the requirements of regulations 14.1 and 14.4 of MARPOL Annex VI.

2.1.2 These guidelines permit two schemes: Scheme A (unit certification with parameter and emission checks and Scheme B (continuous emission monitoring with parameter checks).

2.1.3 For ships which are to use an exhaust gas cleaning system in part or in total in order to comply with regulations 14.1 and/or 14.4 of MARPOL Annex VI, there should be an approved SOX Emissions Compliance Plan (SECP).

2.2 Application

2.2.1 These guidelines apply to any EGC unit as fitted to fuel oil combustion machinery, excluding shipboard incinerators, installed on board a ship.

2.3 Definitions and required documents

Fuel oil
combustion unit
Any en gine, boiler, gas turbine, or other fuel oil fired equipment, excluding shipboard incinerators
EGC Exhaust gas cleaning

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2.3 Definitions and required documents

Fuel oil
combustion unit

Any en gine, boiler, gas turbine, or other fuel oil fired equipment, excluding shipboard incinerators

EGC Exhaust gas cleaning

SOX Sulphur oxides

SO2 Sulphur dioxide

CO2 Carbon dioxide

UTC Universal Time Co-ordinated

Certified Value The SO 2/CO2 ratio specified by the manufacturer that the EGC unit is certified as meeting when operating on a continuous basis on the manufacturers specified maximum fuel sulphur content

In situ Sampling directly within an exhaust gas stream

MCR Maximum Continuous Rating

Load range Maximum rated power of diesel engine or maximum steaming rate of the boiler

SECP SOX Emissions Compliance Plan

SECC SOX Emissions Compliance Certificate

ETM-A EGC system Technical Manual for Scheme A

ETM-B EGC system Technical Manual for Scheme B

OMM Onboard Monitoring Manual

EGC Record Book

A record of the EGC unit in -service operating parameters, component adjustments, maintenance and service records as appropriate